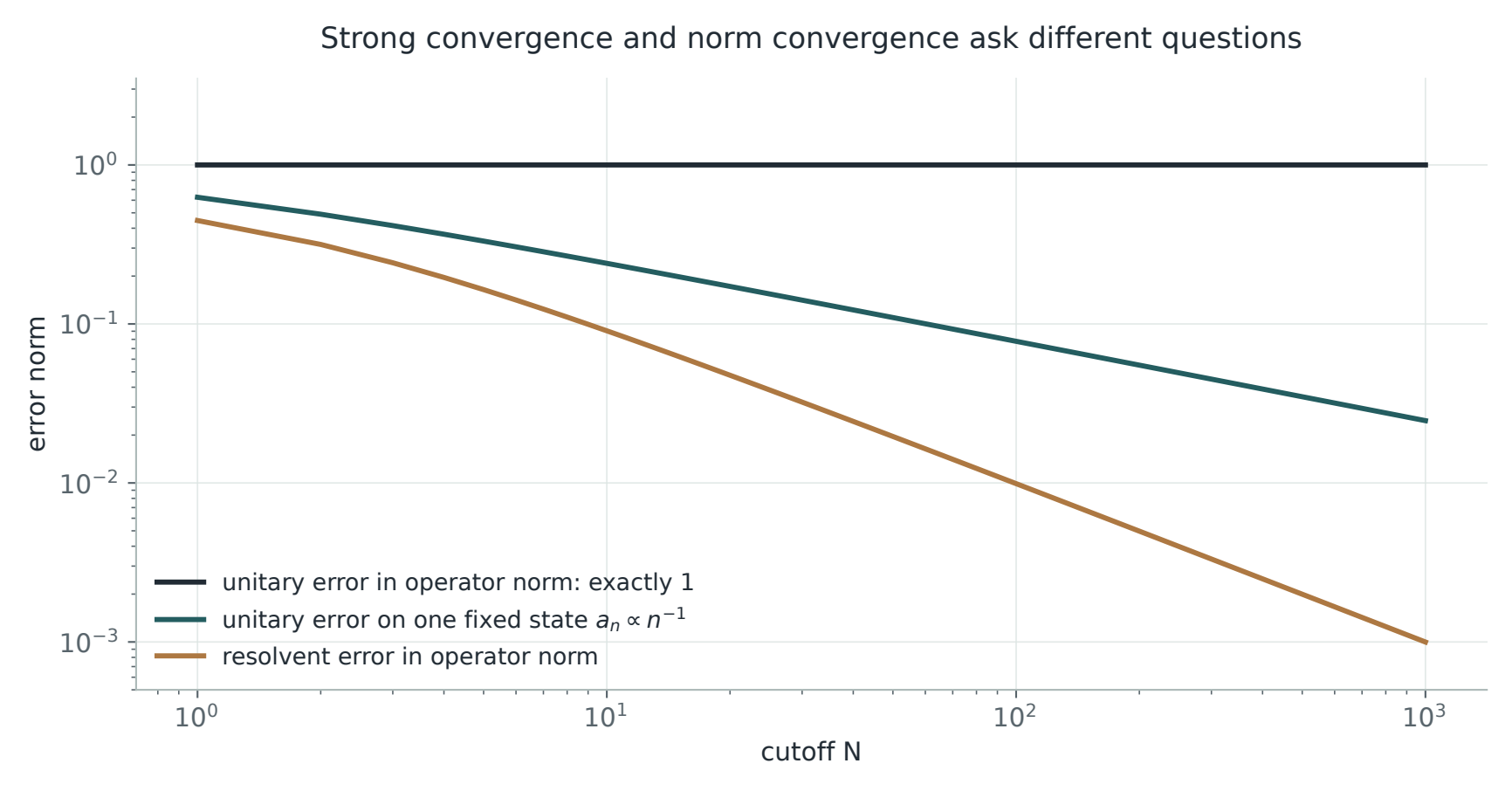
Strong convergence and norm convergence ask different questions
error norm
10^{0}
10^{-1}
10^{-2}
10^{-3}
unitary error in operator norm: exactly 1
unitary error on one fixed state $a_n \propto n^{-1}$
resolvent error in operator norm
10^{0}
10^{1}
10^{2}
10^{3}
cutoff N